NTNU
Faculty of Information Technology
and Electrical Engineering
Department of Electronic Systems

Solutions Exam H21

The solutions to the problems that were not auto-corrected will follow after the pdf included from Inspera.

TFE4146 fall 2021

1 Bonding

Which kind of bonding is found in the diamond lattice elemental semiconductors:
(correct: 2p, wrong: 0p, blank: 0p)

- ○ Ionic bonding
- ○ Van der Waals bonding
- ○ Covalent bonding ✔
- ○ Metallic bonding

Maximum marks: 2

2 Doping

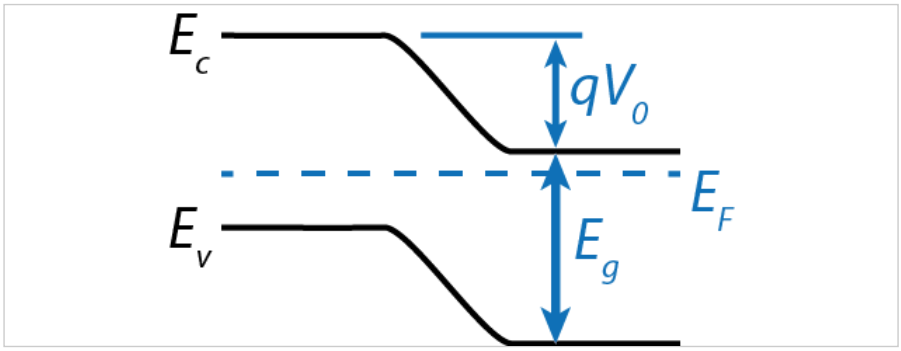
For a sample of Si doped with arsenic to $N_D = 10^{17} cm^{-3}$ in thermal equilibrium at room temperature ($300K$), how far above the intrinsic Fermi level, E_i, is the Fermi level in this material?
Given: $n_i = 1.5 \cdot 10^{10} cm^{-3}$

Fill in your numeric answer here: $E_F - E_i =$ [ ] (0.39 - 0.41)eV.

Maximum marks: 5

1/18

3 p-n junction 1



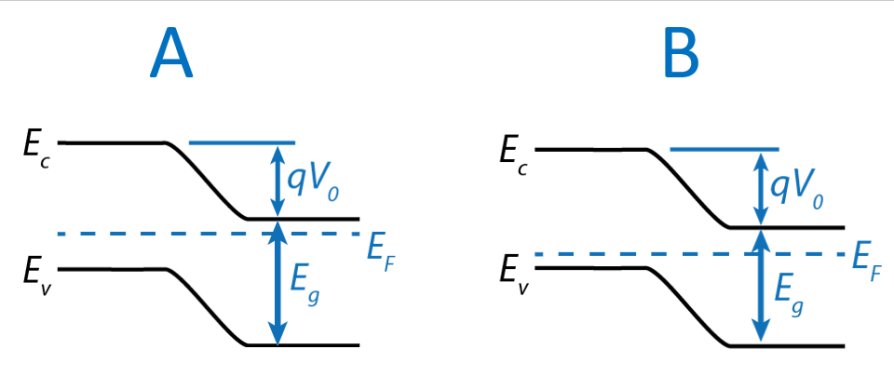
In the following we will consider a Si p-n junction at room temperature. The band diagram at thermal equilibrium for this junction is shown above.

Fill in the blanks in the following two sections of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -0.5p, blank: -0.5p, minimum score: 0p)

Within the depletion region in the p-type material, there is a net Select alternative (positive, neutral, negative)charge. This region is depleted of free Select alternative (donors, acceptors, impurities, carriers)and the remaining charge originates from the ionized Select alternative (donors, acceptors, electrons, holes). Moving further away from the junction into the neutral p-type material, we find a large number of free Select alternative (electrons, holes, dopants) and Select alternative (ionized, neutral, positive) acceptors. The concentration of charge carriers is Select alternative (less than, larger than, essentially equal to)that of the acceptors.

In the n-type material, there are electrons and Select alternative (neutral, very few, free, ionized) donors. In the neutral region, the Select alternative (free, entangled, immobile) electrons and the Select alternative (neutral and free, neutral and immobile, positive and free, positive and immobile) donors cancel out with respect to charge. In the depleted part of the n-type semiconductor, close to the junction, the semiconductor has a net Select alternative (negative, positive, neutral) charge. Here, the semiconductor is depleted of Select alternative (carriers, donors, acceptors) and the net charge is due to Select alternative (holes, unionized donors, ionized donors, ionized acceptors). In this particular p-n junction, the doping is highest Select alternative (at the gym, on the n-side, in the middle, on the p-side). As a result, the current at the junction will be dominated by Select alternative (donors, electrons, acceptors, holes) when the device is under forward bias.

4 Minority carrier concentration

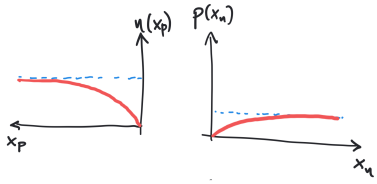


Two p-n junctions, A and B, are shown above. Below are qualitative sketches of the minority carrier concentrations under forward and reverse bias for these junctions. Match the correct label to the correct schematic. *(correct: 1p, wrong: -1p, blank: -1p, minimum score 0p)*

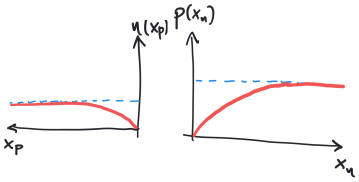
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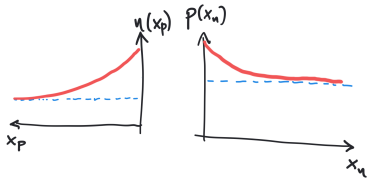
Help

A - reverse ✔ 3

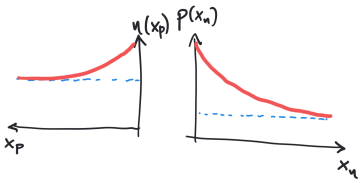


B - reverse ✔ 3





B - forward ✔ 3



A - forward ✔ 3

A - reverse bias | A - forward bias | B - forward bias | B - reverse bias

Maximum marks: 4

5/18

5 Minority carriers

As long as the minority carrier distributions $\delta p(x_n)$ and $\delta n(x_p)$ in a p-n junction are larger than zero, the voltage across the junction will be:
(correct: 2p, wrong: 0p, blank: 0p)

- small and in the forward direction. ✔
- large and in the reverse direction.
- small and in the reverse direction.
- directed directly into k-space.

Maximum marks: 2

6 Diode current 1

Consider a one-sided p+-n junction with a forward bias voltage $V_b >> \frac{kT}{q}$. What happens to the different current components if the bias voltage is increased by a factor 2?
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)
Select one or more alternatives:

- The drift of holes and electrons increase by a factor 2
- The drift of holes (electrons) increase (decrease) by a factor 2
- The diffusion of holes and electrons increase exponentially ✔
- The drift of holes and electrons remain approximately unchanged ✔
- The diffusion of holes (electrons) increase (decrease) exponentially
- The diffusion of holes and electrons increase by a factor 2
- The diffusion of holes increase exponentially, the electron current remain unchanged

Maximum marks: 4

7 Revese breakdown 1

Beyond some critical reverse bias a p-n junction will display "reverse breakdown". The table below lists some keywords that are suitable to describe the two mechanisms for reverse breakdown, "avalance breakdown" and "Zener breakdown".
(correct: 1p, wrong: -1p, blank: -1p, minimum score 0p)

Please match the keywords to the mechanism(s) it is most suitable for.:

	Zener Breakdown	Avalanche breakdown	Both mechanisms
impact ionization	○	○ ✔	○
quantum tunneling	○ ✔	○	○
heavy doping	○ ✔	○	○
reversible / non-destructive	○	○	○ ✔
large E-field	○	○ ✔	○

Maximum marks: 5

8 Reverse breakdown 2

In a so-called P-I-N diode there is a wide intrinsic region between the p- and n-type regions. The p- and n-type regions are typically heavily doped.
(correct: 2p, wrong: 0p, blank: 0p)
Which breakdown mechanism is most relevant for a P-I-N diode with heavily doped p- and n-regions:

- ○ Avalanche breakdown ✔
- ○ Zener breakdown

Maximum marks: 2

9 Reverse breakdown 3

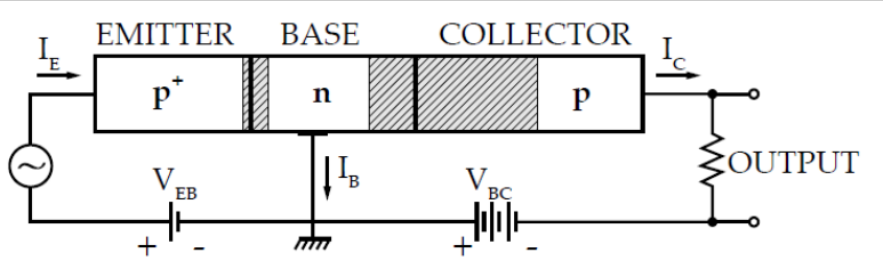
At high electric fields, electrons gain so much energy from the applied electric field that when electrons scatter off each other, they can be scattered from the valence band into the conduction band, this process is known as impact ionization. To be able to scatter an electron into the conduction band, the kinetic energy of the electrons must be greater than the energy gap, E_g. The expression, $E_{kin} = \frac{1}{2}mv_d^2$, can be used to estimate the kinetic energy of an electron with drift velocity, v_d.

A given semiconductor has a band gap of 1.3 eV, an effective mass of 1.1 m_0, and a mobility of 1000 cm²/Vs. Use the condition above to estimate the electric field where impact ionization will take place.

Enter your answer here: (6300000 - 6600000) V/m.

Maximum marks: 5

10 BJT 1



For the BJT shown above, which relationship between the doping concentrations agrees with the sketch:
(correct: 2p, wrong: 0p, blank: 0p)

- $N_E > N_B > N_C$ ✔
- $N_E < N_B < N_C$
- $N_E > N_B < N_C$
- $N_E < N_B > N_C$

What is the operation mode of the BJT shown above:

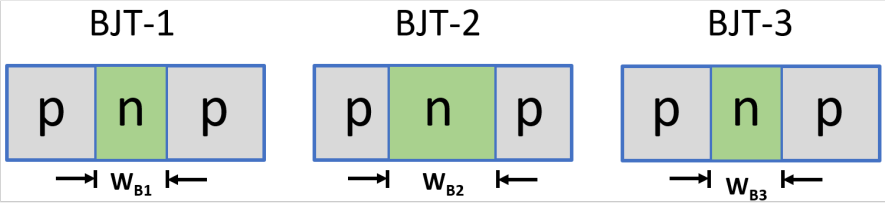
- Saturation
- Cutoff
- Normal active ✔
- Inverted

What is the the biasing arrangement shown above called:

- Common base ✔
- Gummel-Poon
- Common emitter
- Coupled-diode

Maximum marks: 6

11 BJT-3



The three p-n-p bipolar junction transistors schematically illustrated above, are identical except that BJT-2 has a base region that is twice as wide as that of BJT-1 and BJT-3. We also have that BJT-3 and BJT-2 have a doping in the base region that is twice as large as that of BJT-1. That is:

$W_{B2} = 2W_{B1} = 2W_{B3}$ and $N_{B3} = N_{B2} = 2N_{B1}$

We assume that $N_E > N_B > N_C$ for all three transistors.
(correct: 1p, wrong: -1p, blank: -1p, minimum score 0p)

Which of the three transistors will have the highest value for:

	BJT-2	BJT-3	BJT-1
"Punch-through" voltage	✔		
Emitter injection efficiency			✔
Common emitter gain			✔
Base transport factor			✔

Maximum marks: 4

12 P+-N junction (num)

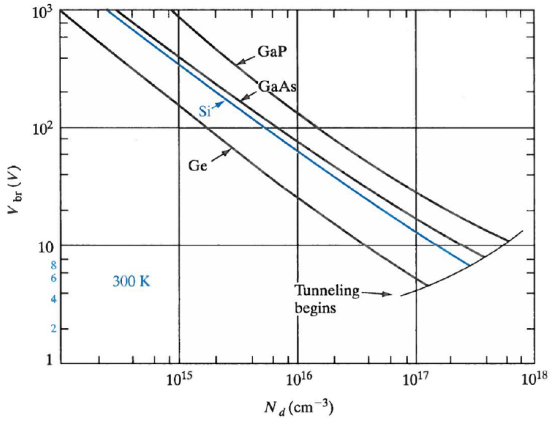


Figure Variation of avalanche breakdown voltage in abrupt p^+-n junctions, as a function of the n-side dopant concentration.

A Si p^+-n junction, 10^{-2} cm^2 in area, has a dopant concentration on the n-side of $N_d=10^{15}cm^{-3}$. The minority carrier lifetime on the n-side is 1µs and the diffusion constant is D_p= 10 cm^2/s. The device is at room temperature (300K). The band gap of Si is 1.1 eV.

a) Calculate the current in units µA for a forward bias of 0.5V.

Enter your numerical answer here: (250 - 300)µA.

b) Approximate the value of the contact potential, qV_0, in units eV for this device.

Enter your numerical answer here: (0.82 - 0.85)eV.

Hint: It may be helpful to sketch the band diagram for this device to make the correct assumptions in your calculations.

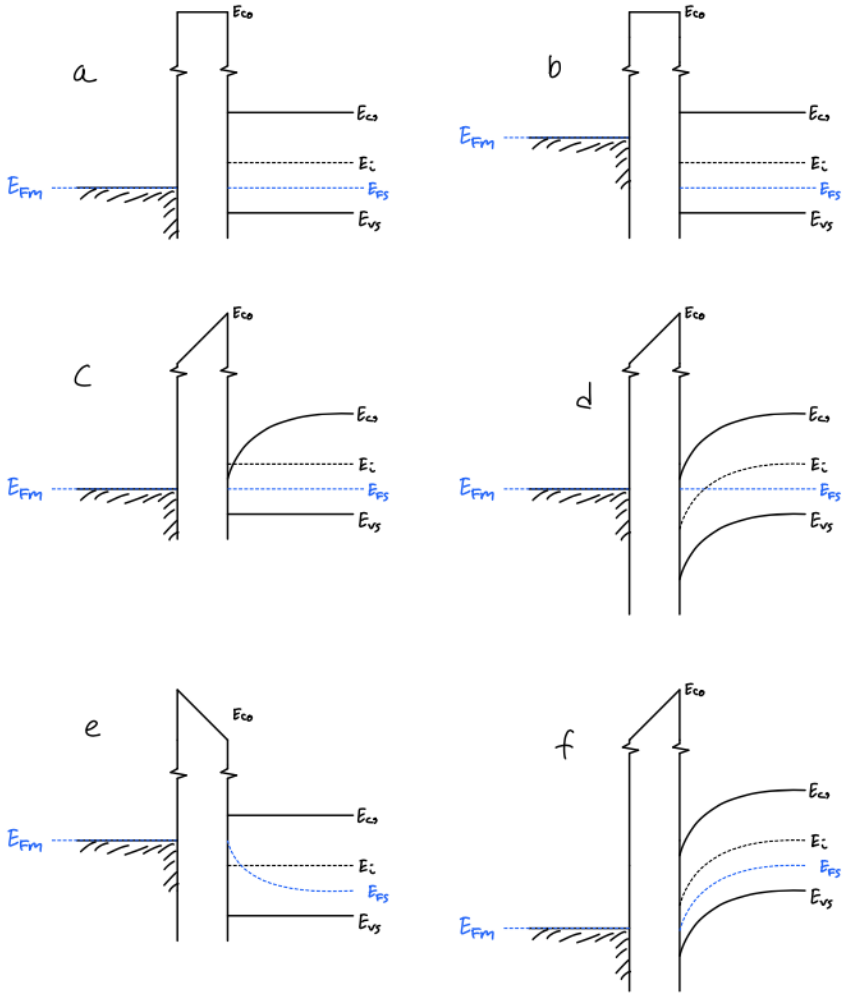
c) Given the figure above, what is the depletion region width W in units µm at the onset of avalanche breakdown in this junction?
(an approximate value from the figure above is sufficient to arrive at an acceptable answer)

Enter your numerical answer here: (18 - 24)µm

Maximum marks: 15

13 MOS Capacitor band structure

The figure shows different illustrations of the band diagram of a MOS structure. Which of the illustrations shows a plausible band diagram for a real MOS capacitor?
(correct: 1p, wrong: -1p, blank: 0p, minimum score 0p)



Select one or more alternatives:

- [ ] c
- [ ] e
- [ ] d ✔
- [ ] f
- [ ] a ✔
- [ ] b ✔

Maximum marks: 3

14 MOS Capacitor bias

Fill in the blanks.
(correct: 2p, wrong: -1p, blank: -1p, minimum score 0p)

Consider an ideal MOS capacitor with a p-type substrate, i.e. flat band voltage is zero. When the gate voltage is increased such that the MOS capacitor is in depletion, the induced charge in the semiconductor is dominated by Select alternative (Ionized doping atoms, Minority carriers, Free electrons in the conduction band, Free holes in the valence band). In this region, the capacitance Select alternative (depends on the frequency, is close to its maximum, is close to its minimum) and the space charge in the semiconductor increase Select alternative (with the square root of, exponentially with) the applied gate voltage.

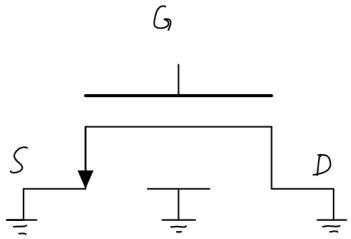
Maximum marks: 6

15 MOS capacitor small-signal

You measure the small-signal capacitance of a MOS capacitor biased in inversion. Which statement is true?
(correct: 2p, wrong: 0p, blank: 0p)
Select one alternative:

- We observe a small capacitance if the frequency is higher than the dielectric relaxation time of the of the substrate.
- We observe a small capacitance if the frequency is low compared to the generation-recombination rate in the inversion layer.
- We observe a large capacitance if the frequency is low compared to the generation recombination rate in the inversion layer. ✔
- We observe a large capacitance if the frequency is higher than the dielectric relaxation time of the of the substrate.

You want to realize an RC low-pass filter, but you do not have any capacitors available. You decide to use a MOSFET as a grounded capacitor by connecting the source, drain and bulk terminal to ground, as shown in the figure below. In a MOSFET, charge can flow very fast into the inversion layer from the source and drain terminal. Assume the MOSFET capacitor is biased in inversion. Would you expect the high-frequency capacitance to be smaller, larger or similar to to that of a MOS capacitor (also biased in inversion)?



Select one alternative

- Similar
- Larger ✔
- Smaller

Maximum marks: 4

16 Current components

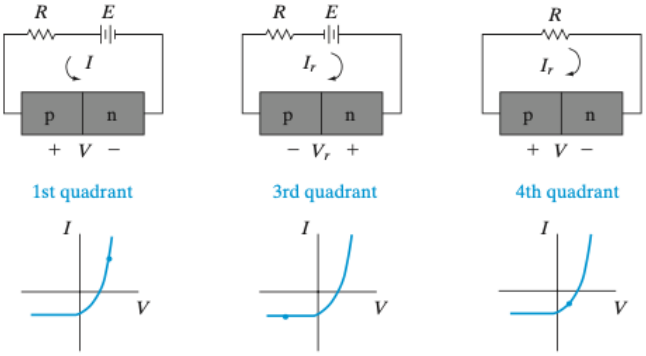
Which of the following currents are dominated by diffusion?
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)
Select one or more alternatives:

- ☐ The drain current in a MOSFET
- ☐ The majority carrier current in the neutral region of a p-n junction
- ☐ Electron-hole-pairs, optically generated within the depletion region of a photodiode, being separated by the junction field
- ☐ The minority carrier current in the neutral region of a p-n junction ✔
- ☐ The hole current in the base region of a p-n-p BJT ✔

Maximum marks: 4

17 Optoelectronic 1

The figure below shows a p-n junction operated in the various quadrants of its I-V characteristic.



(correct: 1p, wrong: -1p, blank: -1p, minimum score 0p)

Select which quadrant the junction should be operated in, when it is to be used as a solar cell, photodetector and a LASER.

	3rd quadrant	1st quadrant	4th quadrant
Photodetector	✔		
Solar cell			✔
LASER		✔	

Maximum marks: 3

18 Photo Diodes

Explain and compare some of the trade-offs related to the design of a solar cell and a photodetector that is to be used in high-speed optical communication.

Maximum marks: 10

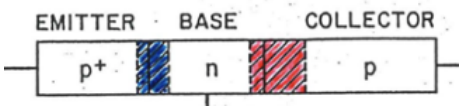
19 Semiconductor LASER

Explain how a semiconductor can be used as a laser. Your answer should include a description of the necessary requirements for stimulated emission and how these conditions can be achieved with a p-n junction.

Maximum marks: 10

20 BJT Doping

Consider the BJT shown below. Determine the relative doping levels in the different regions. Explain the motivation behind these relative doping levels when the transistor is designed for a high common-emitter current gain, β.



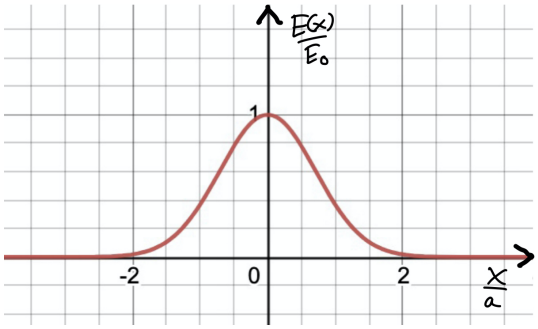
Fill in your answer here

Maximum marks: 10

21 p-n junction (scan)

Consider a p-n junction in thermal equilibrium. The electrical field is given by $E(x) = E_0 \cdot \exp(-\frac{x^2}{a^2})$ as illustrated by the graph below, where E_0 is the maximum absolute value of the electric field and a is some positive constant.

Given: $\int_{-\infty}^{\infty} \exp(-\frac{x^2}{a^2}) = |a|\sqrt{\pi}$



Problem 1 (7p.)
a) Draw a qualitative sketch of the potential $V(x)$ as well as the band diagram. Indicate an appropriate position of the Fermi level in the band diagram.

b) Determine the contact potential V_0.
Hint: You may assume that the p-n junction is infinitely long for this calculation.

Problem 2 (5p.)
Calculate and sketch the space charge density $\rho(x)$. You may use a graphical tool like Desmos or WolframAlpha to help you visualize the function.

Problem 3 (8p.)
a) Sketch a plausible profile for the doping concentration difference $N_d - N_a$.

b) Discuss the accuracy of the depletion approximation for this diode.

c) How would a and E_0 change if the doping concentration is increased on both sides?

Maximum marks: 20

18 Photo Diodes

Assume in the following a solar cell with the structure shown in figure 1.

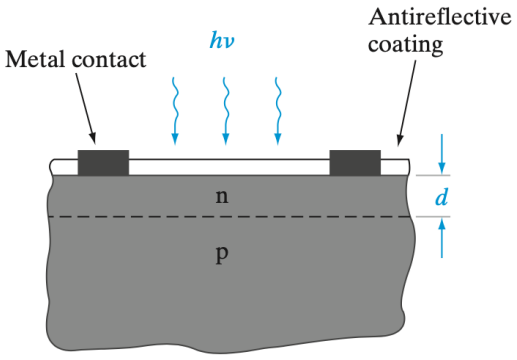


Figure 1: The planar structure of a solar cell

To get the most electrical power from a solar cell, we need

1. As many optically generated electron-hole pairs as possible
2. Collect as many EHPs as possible, by having them diffuse across the depletion region
3. A large contact potential to get a higher energy per EHP collected
4. Low series resistance to prevent ohmic power loss.

For 1. we need a large surface area and a coating that prevent reflection, such that most of the incident light travels into the semiconductor. A trade-off exist in the design of metal contacts to extract the optical current. If the contacts are made at the edges to minimize the interference with the incoming light, a large resistance would result due to the thin n-region near the surface. A usual trade-off between resistance and interference is to lay out the contacts as narrow fingers distributed over the surface.

Another trade-off exist between 2. and 3. To collect most of the generated EHPs, long carrier lifetimes are required. However, increasing the contact potential requires increased doping concentration, which again reduces the carrier lifetime.

While a solar cell is designed to generate power, a photodetector is designed to respond to an incoming optical communication signal. The important parameters of a photodetector are therefore sensitivity, noise and bandwidth. The trade-off related to positioning of the metal contacts would also be present in a photodetector. Covering the detector surface with metal contact for low resistance would decrease the sensitivity, while the high resistance obtained by placing the contacts at the edges would result in a higher RC time-constant and reduced bandwidth. However, there is no need to design for a higher contact potential in a photodetector, as it is the generated current, not the power that is of interest.

For high bandwidth, the photodetector should be designed such that most of the carriers are generated within the depletion region, such that the diffusion time is eliminated. A wider de-

pletion region would increase the sensitivity, but a trade-off must be made as a wider depletion region increase the drift time and therefore reduce bandwidth.

19 Semiconductor Laser

A laser is a device that emits monochromatic, coherent light. In other words, (most of) the photons emitted have the same energy and the same phase. This is achieved by a phenomena called *stimulated emission*. The requirements for stimulated emission are treated in details in chapter 8.3, a short summary is given here.

In contrast to spontaneous emission, stimulated emission occur when an electron in a higher state is stimulated to fall to a lower energy level and emit a photon. This stimulation is provided by the presence of photons whose energy match the energy difference between the two levels. In this setting, some photons will be absorbed, some photons will be emitted by spontaneous emission and some photons will be emitted by stimulated emission. For a good laser, the rate of stimulated emission should be much higher than both the absorption rate and the rate of spontaneous emission (to obtain monochromatic, coherent light).

The stimulated emission is made stronger than the spontaneous emission by increasing the energy density of the photon field. To make the stimulated emission rate higher than the absorption rate, we need *population inversion*, which means that the concentration of electrons is higher in the highest energy level.

These condition can be met in a semiconductor p-n junction as follows. A strong stimulating photon field is achieved by placing the diode in an *optical resonant cavity*. This cavity is made such that the reflection coefficient is smaller than unity in the desired direction of emission, such that a fraction of the internal photons leak out of the cavity.

In a semiconductor, population inversion is achieved when $F_n - F_p > E_g$, which can be achieved by heavy doping and strong forward bias, as shown in figure 2.

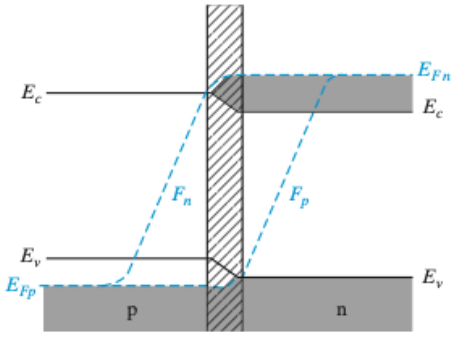


Figure 2: Band diagram of a p-n junction laser under forward bias. The crosshatched region indicates the inversion region at the junction.

20 BJT Doping

Based on the how the width depletion region is distributed between the different regions of the BJT, we see that the relative doping levels are

$$N_E > N_B > N_C. \tag{1}$$

The common-emitter current gain is given by

$$\beta = \frac{i_C}{i_B} = \frac{Bi_{Ep}}{i_{En} + (1-B)i_{Ep}}, \tag{2}$$

where

$$B = \frac{i_C}{i_{Ep}} \tag{3}$$

base transport factor.

Therefore, to have a good p-n-p transistor (high β) we want

1. the emitter current to be composed mainly of holes injected into the base
2. as much as possible of the holes injected by the emitter to be collected
3. the base current to mainly resupply electrons lost by recombination in the base

1. Is achieved by doping the emitter much stronger than the base. This also reduce the part of the base current that contributes to the current across the emitter junction. A weak base doping also increase the base carrier life-time, which in turn increase the base transport factor. A weak collector doping reduces the reverse bias current and increase the avalanche breakdown voltage of the collector junction.

21 P-N Junction

Problem 1

a) The potential $V(x)$ is given by

$$V(x) = V(-\infty) + \int_{-\infty}^{x} -E_0 e^{-\frac{s^2}{a^2}} ds \propto -\text{erf}\left(\frac{x}{a}\right), \tag{4}$$

which is plotted together with the banddiagram in figure 3. Important points:

- $E_c(x), E_v(x) \propto -V(x)$
- Fermi level equally close to conduction band on n-side as to valence band on p-side

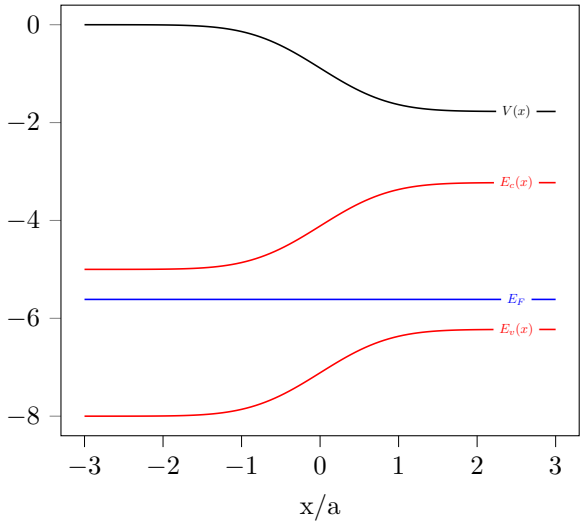


Figure 3: Sketch of potential and banddiagram.

b) The contact potential V_0 is normally found by integrating the electric field across the depletion region. As this p-n junction has an exponentially decaying E-field, there is no well defined depletion region and we need to integrate across the whole device. By assuming the device is infinitely long we are allowed to use the given integral identity. Therefore,

$$V_0 = \int_{-\infty}^{\infty} -E_0 e^{-\frac{x^2}{a^2}} dx = -E_0 a\sqrt{\pi}, \tag{5}$$

where the negative sign means that the potential is decaying in the positive x-direction.

Problem 2

The relation between electric field and space charge density is given by Gauss' law as

$$\rho(x) = \epsilon \frac{dE(x)}{dx} = -2\epsilon E_0 \frac{2x}{a^2} e^{-\frac{x^2}{a^2}}. \tag{6}$$

The space charge density is plotted in figure 4.

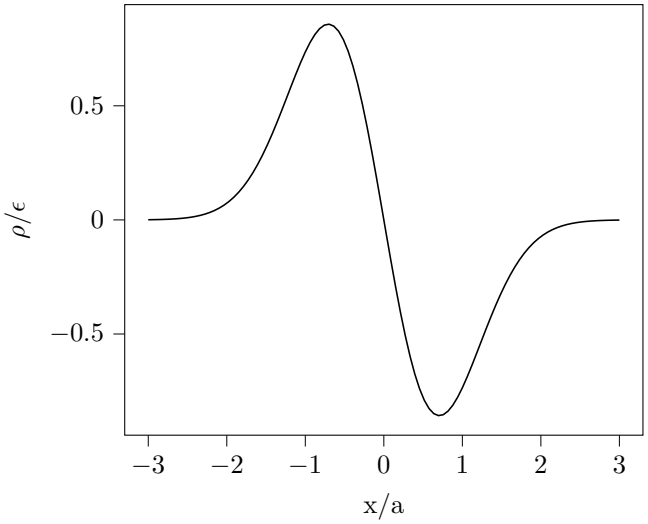


Figure 4: Space charge density ρ, normalized to ϵ

Problem 3

a) The space charge density $\rho(x)$ is caused by ionized doping atoms. If complete ionization is assumed, the doping profile is related to ρ as

$$(N_d - N_a)(x) = \frac{\rho(x)}{q}. \tag{7}$$

However, it is evident from the shape of $\rho(x)$ that this diode does not have a well-defined depletion region where complete ionization can be assumed for the whole region. A reasonable approach could be to assume complete ionization close to the junction and that the fraction of ionized doping atoms is gradually reduced with the distance to the center. This assumption would lead to something similar to the profile shown in figure 5. The important points in this sketch is that $\rho(x)$ follows $(N_d - N_a)$ for $|x|$ close to zero and that $N_d - N_a$ approaches some constant value further away from the junction.

b) The depletion approximation is the approximation that all doping atoms are ionized within the depletion region, and that no E-field exist outside it. Due to the gradual decay in space charge, as shown in figure 4, there is no obvious place to draw a boundary of such an ideal depletion region.

However, a reasonable mathematical approximation could be made by creating an equivalent distribution $\hat{\rho}(x)$ such that

$$\hat{\rho}(x) = \begin{cases} q(N_d - N_a)(x) & \text{if } |x| < c \\ 0 & \text{otherwise,} \end{cases} \tag{8}$$

for some positive constant c which is chosen such that the resulting E-field $\hat{E}(x)$ has the same maximum value $\hat{E}(x = 0) = E_0$ (i.e. the two space charge distributions contains the same amount of charge on each side of the junction).

From figure 6 we observe that this equivalent space charge distribution gives a reasonably accurate approximation for the E-field close to the junction.

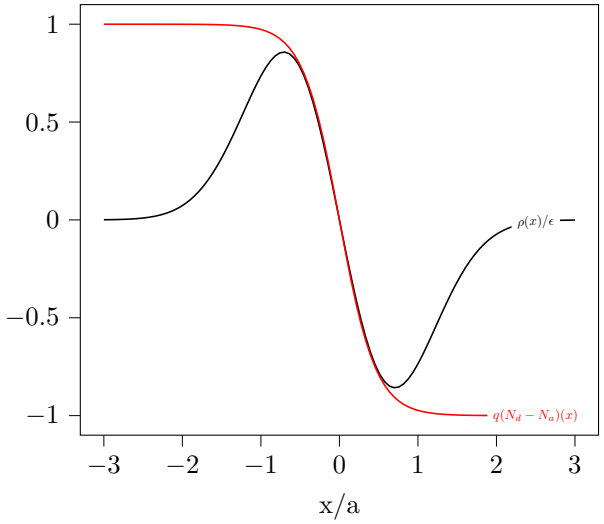


Figure 5: A plausible profile of $N_d - N_a$, plotted together with the space charge density distribution.

c) As the space charge density is proportional to the doping concentration, increasing the doping concentration would increase the peak space charge density. As a consequence, E_0 will increase and a would decrease

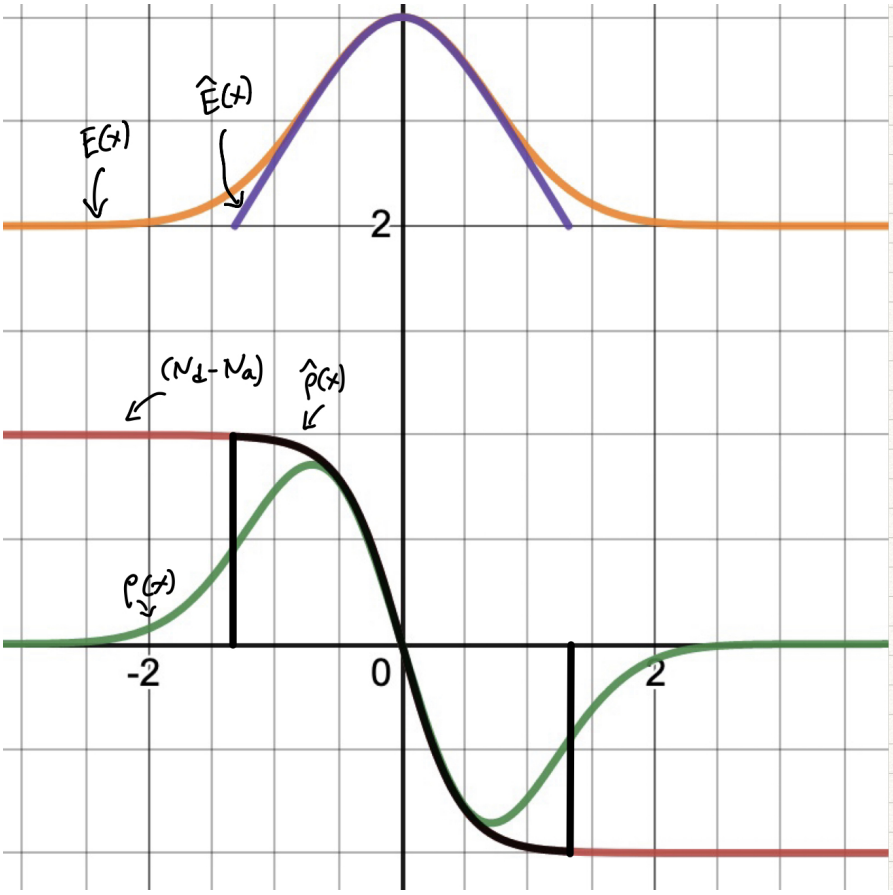


Figure 6: A sketch of an alternative space charge density distribution $\hat{\rho}(x)$ which is chosen such $\hat{\rho}(x) = q(N_d - N_a)$ for $|x| < c$. The constant c is such that the resulting electric field $\hat{E}(x)$ has the same maximum E_0. The doping profile is included for reference, but not drawn to scale.